EXPERIMENT 5

Real-Time Streaming Data Processing using Azure Databricks and Azure Event Hub

Aim

To process real-time streaming data using Azure Databricks and Azure Event Hub by ingesting a live data stream, processing it using Structured Streaming in PySpark, and displaying the output in a Databricks notebook.

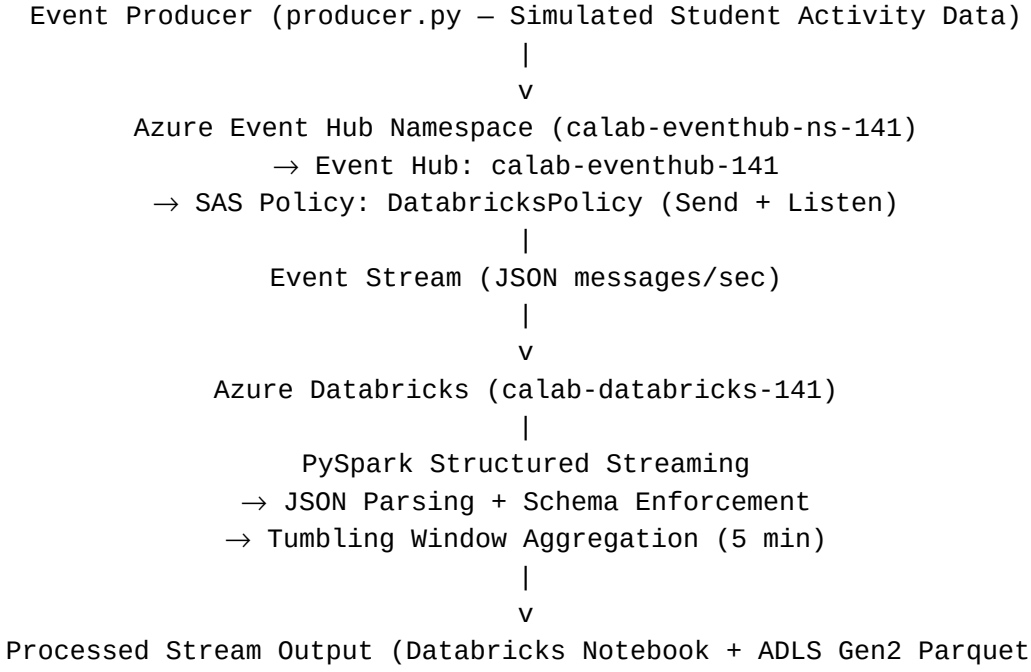
Theory

Azure Event Hub is a fully managed, real-time data ingestion service capable of receiving and processing millions of events per second. It acts as the front door of an event pipeline, decoupling event producers from event consumers. Event Hub supports the Apache Kafka protocol and is widely used for telemetry, clickstream, and IoT data ingestion.

Azure Databricks is an Apache Spark-based analytics platform optimized for Azure. It provides a collaborative notebook environment for big data processing and machine learning. Databricks Structured Streaming is a scalable, fault-tolerant stream processing engine built on Spark SQL that enables continuous processing of real-time data with micro-batch execution.

In this experiment, Azure Event Hub acts as the source of real-time streaming data. Azure Databricks reads the stream using the Event Hub connector, processes the incoming messages using PySpark Structured Streaming, and displays the results continuously in the Databricks notebook. Sample streaming data is simulated by sending events from student records including names like Abhi, Kshitiz, Aakriti, Rohit, Sandhya, and Riya.

Flow Diagram



Procedure

1. Create an Azure Event Hub Namespace calab-eventhub-ns-141 in the Azure portal inside resource group 25MCAR0141.
2. Inside the namespace, create an Event Hub named calab-eventhub-141 with 2 partitions.
3. Navigate to Shared Access Policies → add a policy DatabricksPolicy with Send and Listen permissions.
4. Copy the Connection String Primary Key for use in Databricks.
5. Create an Azure Databricks workspace named calab-databricks-141 in the same resource group.
6. Launch Databricks Studio → create a new cluster with single-node configuration (minimum cost).
7. Install the Event Hub connector library: com.microsoft.azure:azure-eventhubs-spark_2.12:2.3.22 via Maven.
8. Create a new Databricks notebook and configure the Event Hub connection string.
9. Write PySpark Structured Streaming code to read from Event Hub as source.
10. Send sample events (student records: Abhi, Kshitiz, Aakriti, Rohit, Sandhya, Riya) to Event Hub using producer.py.
11. Run the streaming notebook and observe real-time events and tumbling window aggregations displayed in the output cell.
12. Stop the stream and terminate the cluster after capturing the output.

Expected Output

<GitHub Link Here>

Result

Thus, real-time streaming data was successfully processed using Azure Databricks and Azure Event Hub. The Event Hub ingested simulated student event data and Databricks Structured Streaming consumed, processed, and displayed the events in real time using PySpark. Tumbling window aggregations provided per-student event counts and average scores. This experiment demonstrates a scalable cloud-native architecture for real-time data pipelines widely used in IoT, fraud detection, and live analytics applications.